# 電子學實驗 (一)
# Integrators, Differentiators, and Their Frequency Responses
# 結報

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# 1 Exp. 1 Integrator

## I Circuit

Measure $v_I$ and $v_O$ waveforms of an op amp where input voltage $v_I$ is connected to $v_-$ through an input resistance $R = 1$ kΩ(current from $v_I$ to $v_-$), $v_+$ is connected to ground, and $v_O$ is fed back to $v_-$ through a capacitance $C = 0.01$ μF (103) (current from $v_-$ to $v_O$), under triangle, square, and sinusoidal $v_I$ with amplitude $0.2$.

### i Theoretical results

$v_O(0^-)$ is the initial voltage across $C$ ($v_O$ relative to $v_-$).

Time domain:

$$v_+ = v_- = 0$$

$$i_I = \frac{v_I}{R} = i_C = -C\frac{\mathrm{d}v_O}{\mathrm{d}t}$$

$$v_O = v_O(0^-) - \frac{1}{RC}\int_0^t v_I(\tau)\,\mathrm{d}\tau$$

Laplace domain:

$$V_+ = V_- = 0$$

$$I_I = \frac{V_I}{R} = I_C = -CsV_O + Cv_O(0^-)$$

$$V_O = \frac{1}{s}v_O(0^-) - \frac{1}{RCs}V_I$$

Transfer function: Infinite DC gain. $90$ degrees phase shift.

$$H(s) = \frac{V_o}{V_i} = -\frac{1}{RCs}$$

Unity-gain frequency

$$\frac{1}{2\pi RC} \approx 15.9 \text{ kHz}$$

Zero-state responses:

- Triangle wave input:

$$v_I(t) = 4A\left(\left|\frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor\right| - \frac{1}{4}\right), \quad A, T > 0$$

$$v_O(t) = -\frac{AT}{RC}\left(2s\,|s| - s\right), \quad s = \frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor$$

- Square wave input:

$$v_I(t) = 2A\left(\left\lfloor\frac{t}{T}\right\rfloor - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor + \frac{1}{2}\right), \quad A, T > 0$$

  Triangle wave output with amplitude $\frac{AT}{4RC}$

$$v_O(t) = -\frac{AT}{RC}\left(\left|\frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor\right| - \frac{1}{4}\right)$$

- Sinusoidal input:

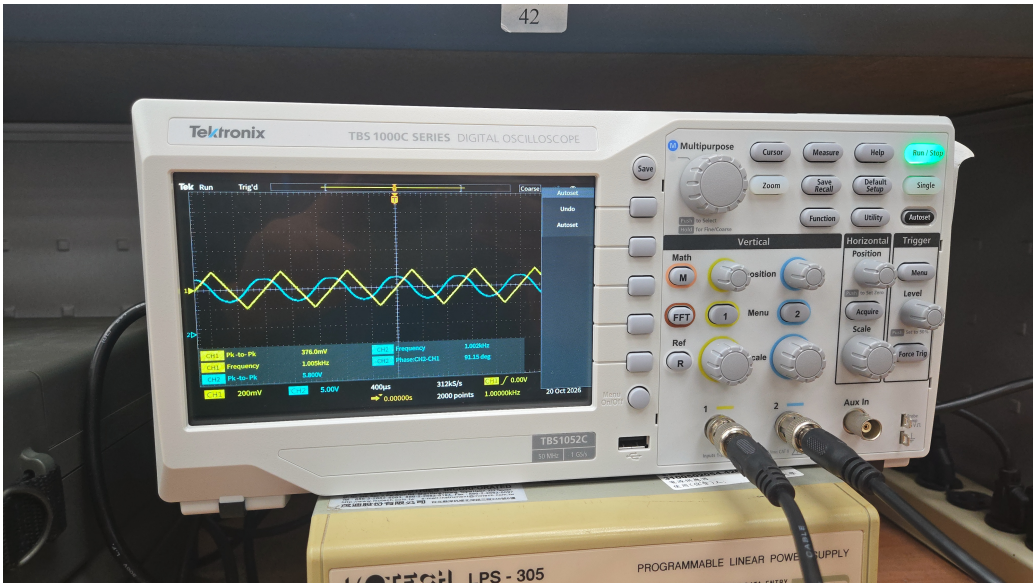
$$v_I(t) = A\sin\left(\frac{t}{T}\right), \quad A, T > 0$$

Sinusoidal output 90 degrees leading input with amplitude $\frac{AT}{RC}$.

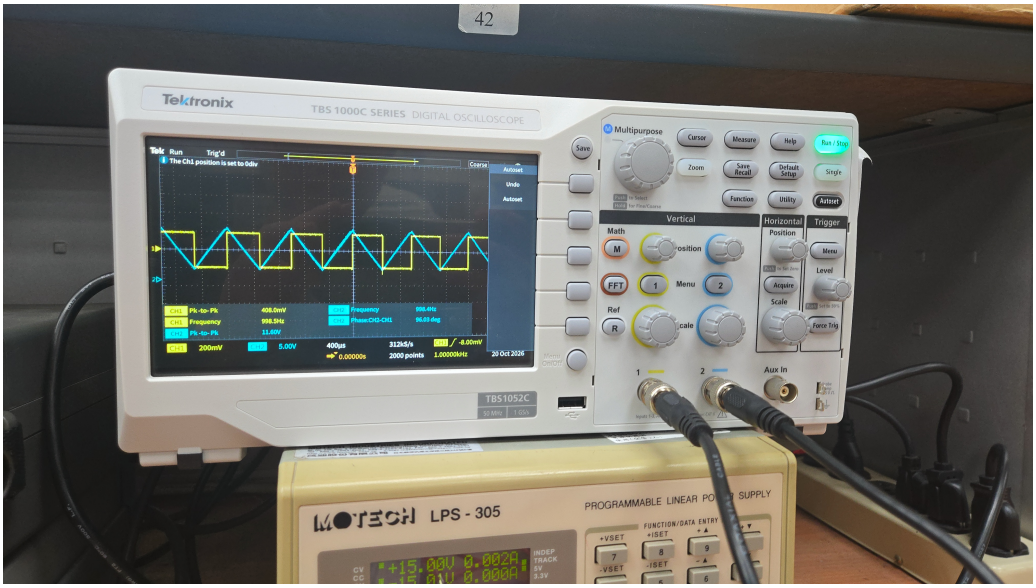
$$v_O(t) = \frac{AT}{RC}\cos\left(\frac{t}{T}\right)$$

## II 實驗結果

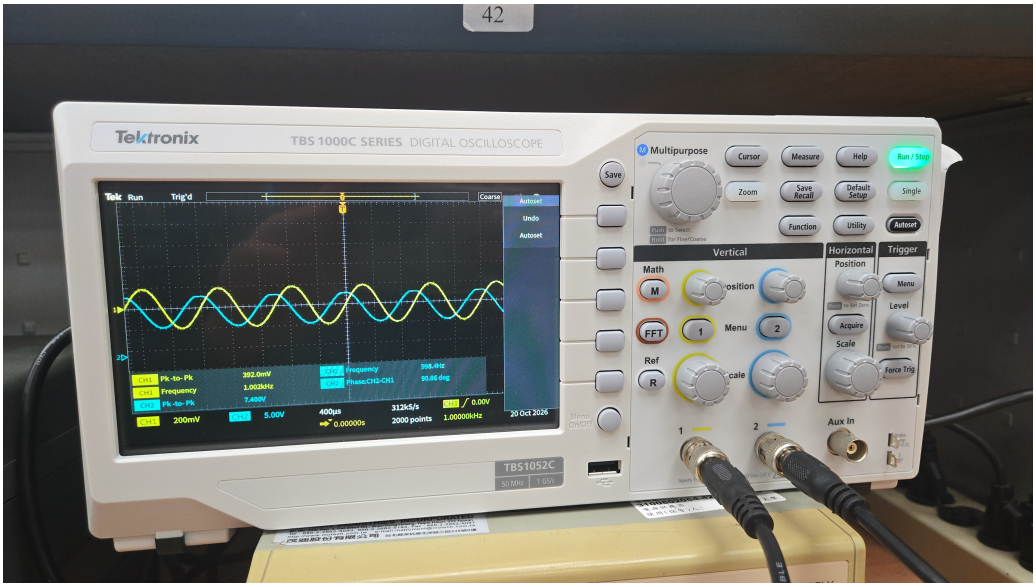
Triangle wave:




Square wave:



Sinusoidal wave:




# 2 Exp. 2 Differentiator

## I Circuit

Measure $v_I$ and $v_O$ waveforms of an op amp where input voltage $v_I$ is connected to $v_-$ through an input capacitor $C = 0.01$ µF (103) (current from $v_I$ to $v_-$), $v_+$ is connected to ground, and $v_O$ is fed back to $v_-$ through a resistor $R = 1$ kΩ(current from $v_-$ to $v_O$), under triangle, square, and sinusoidal $v_I$ with amplitude $0.2$.

## II Theoretical results

$v_I(0^-)$ is the initial voltage across $C$ ($v_I$ relative to $v_-$).

Time domain:

$$i_I = C\frac{\mathrm{d}v_I}{\mathrm{d}t} = i_f = \frac{-v_O}{R}$$

$$v_O = -RC\frac{\mathrm{d}v_I}{\mathrm{d}t}$$

Laplace domain:

$$I_I = CsV_I - Cv_I(0^-) = I_f = \frac{-V_O}{R}$$

$$V_O = -RCsV_I + RCv_I(0^-)$$

Transfer function: Infinite gain at $\omega = \infty$. $-90$ degrees phase shift.

$$H(s) = \frac{V_o}{V_i} = -RCs$$

Unity-gain frequency

$$\frac{1}{2\pi RC} \approx 15.9 \text{ kHz}$$

Responses:

- Triangle wave input:

$$v_I(t) = 4A\left(\left|\frac{t}{T} - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor\right| - \frac{1}{4}\right), \quad A, T > 0$$

  Square wave with amplitude $\frac{4ARC}{T}$

$$v_O(t) = -\frac{8ARC}{T}\left(\left\lfloor\frac{t}{T}\right\rfloor - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor + \frac{1}{2}\right), \quad \frac{2t}{T} \notin \mathbb{Z}$$

- Square wave input:

$$v_I(t) = 2A\left(\left\lfloor\frac{t}{T}\right\rfloor - \left\lfloor\frac{t}{T} + \frac{1}{2}\right\rfloor + \frac{1}{2}\right), \quad A, T > 0$$

  Dirac delta sum output

$$v_O(t) = 2ARC\sum_{n\in\mathbb{Z}} \delta(t - nT) - \delta(t - \left(n + \frac{1}{2}\right)T)$$

- Sinusoidal input:

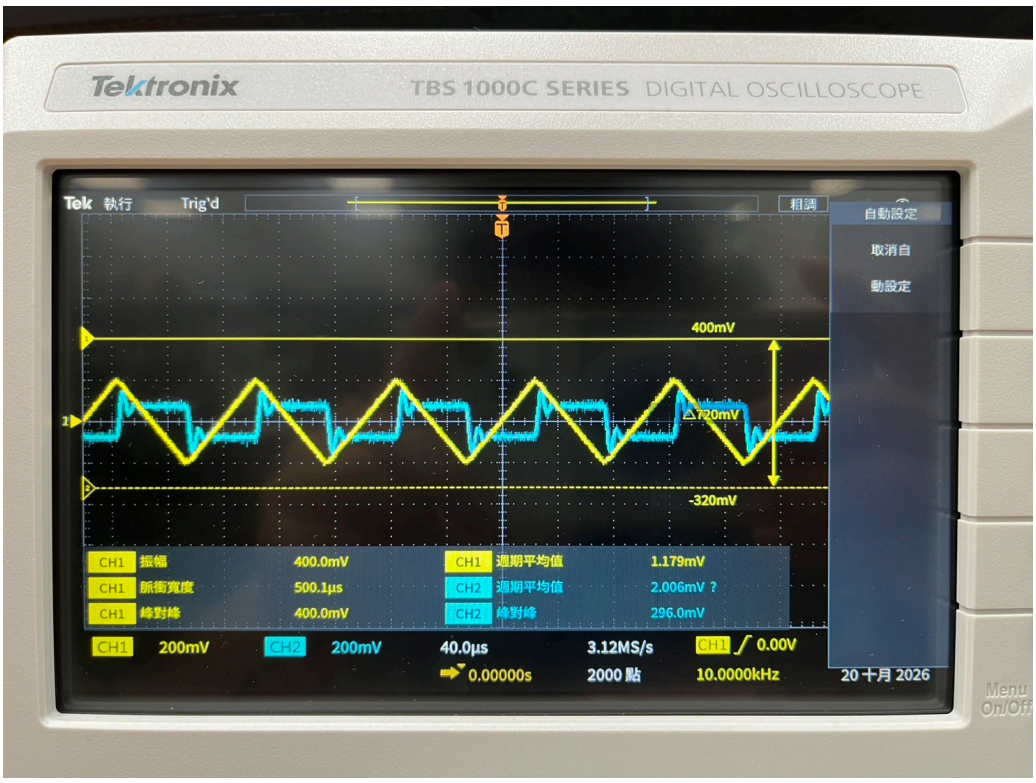
$$v_I(t) = A\sin\left(\frac{t}{T}\right), \quad A, T > 0$$

  Sinusoidal output 90 degrees lagging input with amplitude $\frac{ARC}{T}$.

$$v_O(t) = -\frac{ARC}{T}\cos\left(\frac{t}{T}\right)$$

## III 實驗結果與數據分析

Triangle wave:




Square wave:




Sinusoidal wave:




# 3 Exp. 3 Frequency Responses (I) Integrator

## I Circuit

Measure frequency response of the circuit in Exp. 1 Integrator under

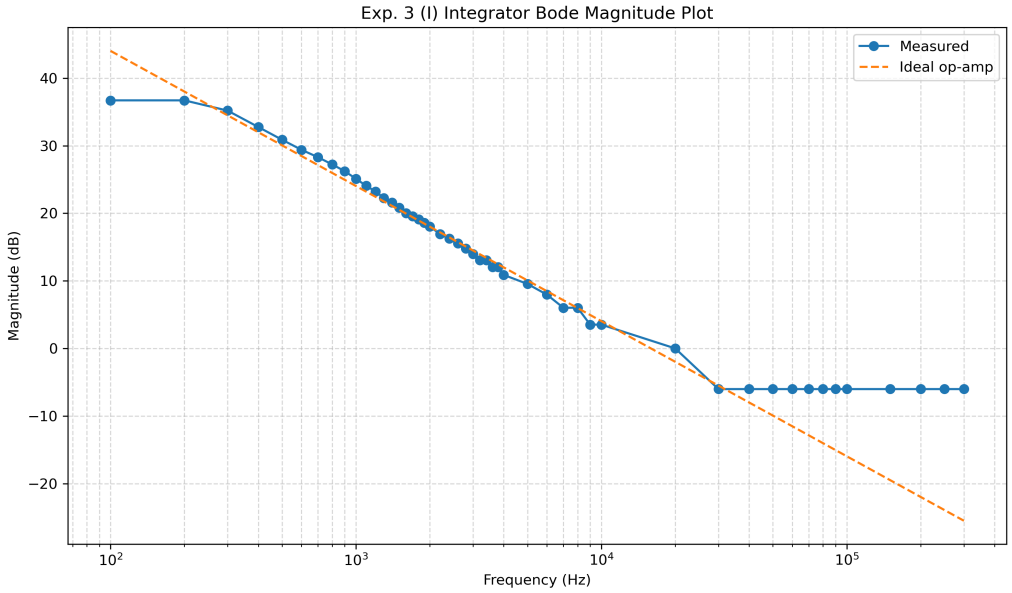
$$v_I = 0.2\sin(2\pi f t), \quad f = 100\text{ Hz} \sim 300\text{ kHz}$$

and sketch the Bode magnitude plot.

# II 實驗結果與數據分析

| frequency (Hz) | measured output voltage peak-to-peak (V) | measured gain (dB) | ideal output voltage peak-to-peak (V) | ideal gain (dB) | error w.r.t. ideal (%) |
|---|---|---|---|---|---|
| 100 | 27.40 | 36.71 | 63.66 | 44.04 | -56.96 |
| 200 | 27.40 | 36.71 | 31.83 | 38.02 | -13.92 |
| 300 | 23.00 | 35.19 | 21.22 | 34.49 | 8.38 |
| 400 | 17.40 | 32.77 | 15.92 | 32.00 | 9.33 |
| 500 | 14.00 | 30.88 | 12.73 | 30.06 | 9.96 |
| 600 | 11.80 | 29.40 | 10.61 | 28.47 | 11.21 |
| 700 | 10.40 | 28.30 | 9.09 | 27.13 | 14.35 |
| 800 | 9.20 | 27.23 | 7.96 | 25.97 | 15.61 |
| 900 | 8.20 | 26.24 | 7.07 | 24.95 | 15.92 |
| 1000 | 7.20 | 25.11 | 6.37 | 24.04 | 13.10 |
| 1100 | 6.40 | 24.08 | 5.79 | 23.21 | 10.58 |
| 1200 | 5.80 | 23.23 | 5.31 | 22.45 | 9.33 |
| 1300 | 5.20 | 22.28 | 4.90 | 21.76 | 6.19 |
| 1400 | 4.80 | 21.58 | 4.55 | 21.11 | 5.56 |
| 1500 | 4.40 | 20.83 | 4.24 | 20.51 | 3.67 |
| 1600 | 4.00 | 20.00 | 3.98 | 19.95 | 0.53 |
| 1700 | 3.80 | 19.55 | 3.74 | 19.43 | 1.47 |
| 1800 | 3.60 | 19.08 | 3.54 | 18.93 | 1.79 |
| 1900 | 3.40 | 18.59 | 3.35 | 18.46 | 1.47 |
| 2000 | 3.20 | 18.06 | 3.18 | 18.02 | 0.53 |
| 2200 | 2.80 | 16.90 | 2.89 | 17.19 | -3.24 |
| 2400 | 2.60 | 16.26 | 2.65 | 16.43 | -1.98 |
| 2600 | 2.40 | 15.56 | 2.45 | 15.74 | -1.98 |
| 2800 | 2.20 | 14.81 | 2.27 | 15.09 | -3.24 |
| 3000 | 2.00 | 13.98 | 2.12 | 14.49 | -5.75 |
| 3200 | 1.80 | 13.06 | 1.99 | 13.93 | -9.52 |
| 3400 | 1.80 | 13.06 | 1.87 | 13.41 | -3.87 |
| 3600 | 1.60 | 12.04 | 1.77 | 12.91 | -9.52 |
| 3800 | 1.60 | 12.04 | 1.68 | 12.44 | -4.50 |
| 4000 | 1.40 | 10.88 | 1.59 | 12.00 | -12.04 |
| 5000 | 1.20 | 9.54 | 1.27 | 10.06 | -5.75 |
| 6000 | 1.00 | 7.96 | 1.06 | 8.47 | -5.75 |
| 7000 | 0.80 | 6.02 | 0.91 | 7.13 | -12.04 |
| 8000 | 0.80 | 6.02 | 0.80 | 5.97 | 0.53 |
| 9000 | 0.60 | 3.52 | 0.71 | 4.95 | -15.18 |
| 10000 | 0.60 | 3.52 | 0.64 | 4.04 | -5.75 |
| 20000 | 0.40 | 0.00 | 0.32 | -1.98 | 25.66 |
| 30000 | 0.20 | -6.02 | 0.21 | -5.51 | -5.75 |
| 40000 | 0.20 | -6.02 | 0.16 | -8.00 | 25.66 |
| 50000 | 0.20 | -6.02 | 0.13 | -9.94 | 57.08 |
| 60000 | 0.20 | -6.02 | 0.11 | -11.53 | 88.50 |
| 70000 | 0.20 | -6.02 | 0.09 | -12.87 | 119.91 |
| 80000 | 0.20 | -6.02 | 0.08 | -14.03 | 151.33 |
| 90000 | 0.20 | -6.02 | 0.07 | -15.05 | 182.74 |
| 100000 | 0.20 | -6.02 | 0.06 | -15.96 | 214.16 |
| 150000 | 0.20 | -6.02 | 0.04 | -19.49 | 371.24 |
| 200000 | 0.20 | -6.02 | 0.03 | -21.98 | 528.32 |
| 250000 | 0.20 | -6.02 | 0.03 | -23.92 | 685.40 |
| 300000 | 0.20 | -6.02 | 0.02 | -25.51 | 842.48 |

## III Bode plot




# 4 Exp. 3 Frequency Responses (I) Differentiator

## I Circuit

Measure frequency response of the circuit in Exp. 2 Differentiator under

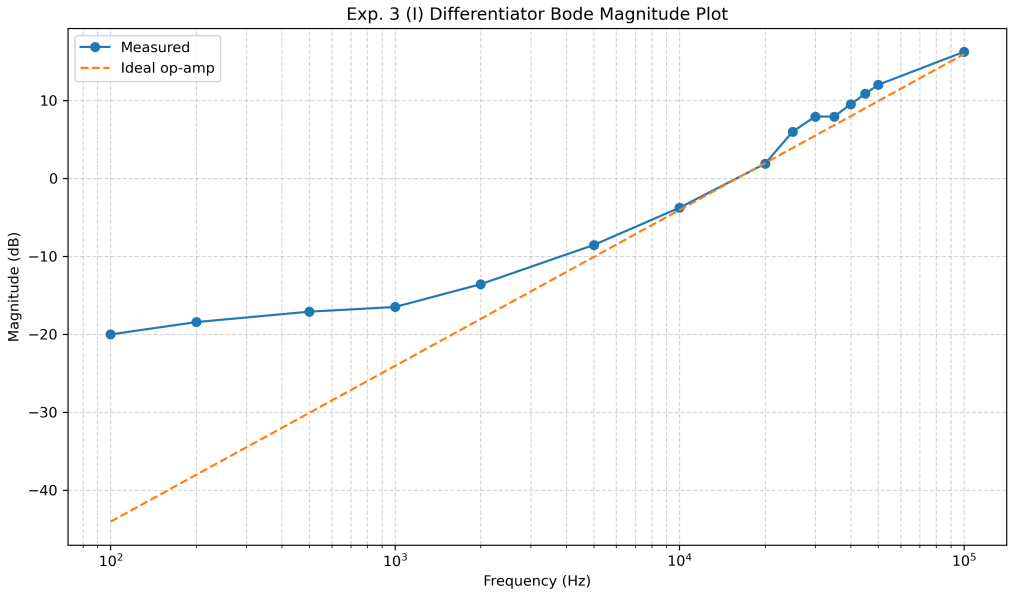
$$v_I = 0.2\sin(2\pi f t), \quad f = 100\text{ Hz} \sim 100\text{ kHz}$$

and sketch the Bode magnitude plot.

## II 實驗結果與數據分析

| frequency (Hz) | measured output voltage peak-to-peak (V) | measured gain (dB) | ideal output voltage peak-to-peak (V) | ideal gain (dB) | error w.r.t. ideal (%) |
|---|---|---|---|---|---|
| 100 | 0.04 | -20.00 | 0.00 | -44.04 | 1491.55 |
| 200 | 0.05 | -18.42 | 0.01 | -38.02 | 854.93 |
| 500 | 0.06 | -17.08 | 0.01 | -30.06 | 345.63 |
| 1000 | 0.06 | -16.48 | 0.03 | -24.04 | 138.73 |
| 2000 | 0.08 | -13.56 | 0.05 | -18.02 | 67.11 |
| 5000 | 0.15 | -8.52 | 0.13 | -10.06 | 19.37 |
| 10000 | 0.26 | -3.74 | 0.25 | -4.04 | 3.45 |
| 20000 | 0.50 | 1.90 | 0.50 | 1.98 | -0.93 |
| 25000 | 0.80 | 6.02 | 0.63 | 3.92 | 27.32 |
| 30000 | 1.00 | 7.96 | 0.75 | 5.51 | 32.63 |
| 35000 | 1.00 | 7.96 | 0.88 | 6.84 | 13.68 |
| 40000 | 1.20 | 9.54 | 1.01 | 8.00 | 19.37 |
| 45000 | 1.40 | 10.88 | 1.13 | 9.03 | 23.79 |
| 50000 | 1.60 | 12.04 | 1.26 | 9.94 | 27.32 |
| 100000 | 2.60 | 16.26 | 2.51 | 15.96 | 3.45 |

## III Bode plot




# 5 Exp. 3 Frequency Responses (II) Integrator without feedback resistor

## I Circuit

Measure frequency response of the circuit in Exp. 1 Integrator with the capacitor $C$ replaced with a capacitor $C = 0.001$ μF (102) under

$$v_I = 0.2 \sin(2\pi f t), \quad f = 100 \text{ Hz} \sim 1 \text{ MHz}$$

and sketch the Bode magnitude plot.

## II Theoretical results

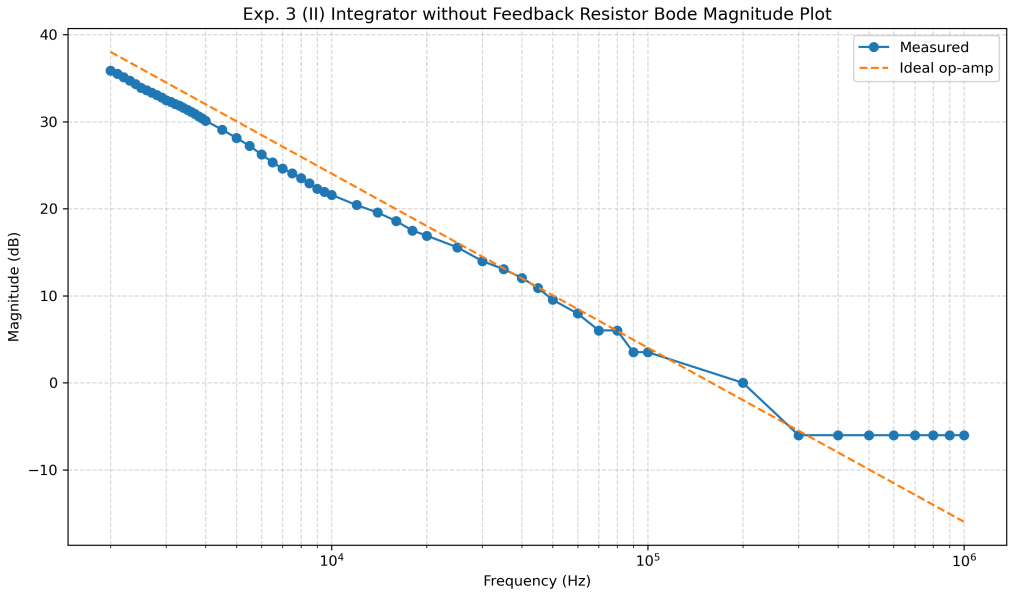
Unity-gain frequency

$$\frac{1}{2\pi RC} \approx 159 \text{ kHz}$$

# III 實驗結果與數據分析

| frequency (Hz) | measured output voltage peak-to-peak (V) | measured gain (dB) | ideal output voltage peak-to-peak (V) | ideal gain (dB) | error w.r.t. ideal (%) |
|---|---|---|---|---|---|
| 2000 | 24.80 | 35.85 | 31.83 | 38.02 | -22.09 |
| 2100 | 23.80 | 35.49 | 30.32 | 37.59 | -21.49 |
| 2200 | 22.80 | 35.12 | 28.94 | 37.19 | -21.21 |
| 2300 | 21.80 | 34.73 | 27.68 | 36.80 | -21.24 |
| 2400 | 20.80 | 34.32 | 26.53 | 36.43 | -21.59 |
| 2500 | 19.80 | 33.89 | 25.46 | 36.08 | -22.25 |
| 2600 | 19.20 | 33.62 | 24.49 | 35.74 | -21.59 |
| 2700 | 18.60 | 33.35 | 23.58 | 35.41 | -21.11 |
| 2800 | 18.00 | 33.06 | 22.74 | 35.09 | -20.83 |
| 2900 | 17.40 | 32.77 | 21.95 | 34.79 | -20.74 |
| 3000 | 16.80 | 32.46 | 21.22 | 34.49 | -20.83 |
| 3100 | 16.40 | 32.26 | 20.54 | 34.21 | -20.14 |
| 3200 | 16.00 | 32.04 | 19.89 | 33.93 | -19.58 |
| 3300 | 15.60 | 31.82 | 19.29 | 33.67 | -19.14 |
| 3400 | 15.20 | 31.60 | 18.72 | 33.41 | -18.82 |
| 3500 | 14.80 | 31.36 | 18.19 | 33.16 | -18.63 |
| 3600 | 14.40 | 31.13 | 17.68 | 32.91 | -18.57 |
| 3700 | 14.00 | 30.88 | 17.21 | 32.67 | -18.63 |
| 3800 | 13.60 | 30.63 | 16.75 | 32.44 | -18.82 |
| 3900 | 13.20 | 30.37 | 16.32 | 32.22 | -19.14 |
| 4000 | 12.80 | 30.10 | 15.92 | 32.00 | -19.58 |
| 4500 | 11.40 | 29.10 | 14.15 | 30.97 | -19.42 |
| 5000 | 10.20 | 28.13 | 12.73 | 30.06 | -19.89 |
| 5500 | 9.20 | 27.23 | 11.57 | 29.23 | -20.52 |
| 6000 | 8.20 | 26.24 | 10.61 | 28.47 | -22.72 |
| 6500 | 7.40 | 25.34 | 9.79 | 27.78 | -24.44 |
| 7000 | 6.80 | 24.61 | 9.09 | 27.13 | -25.23 |
| 7500 | 6.40 | 24.08 | 8.49 | 26.54 | -24.60 |
| 8000 | 6.00 | 23.52 | 7.96 | 25.97 | -24.60 |
| 8500 | 5.60 | 22.92 | 7.49 | 25.45 | -25.23 |
| 9000 | 5.20 | 22.28 | 7.07 | 24.95 | -26.49 |
| 9500 | 5.00 | 21.94 | 6.70 | 24.48 | -25.39 |
| 10000 | 4.80 | 21.58 | 6.37 | 24.04 | -24.60 |
| 12000 | 4.20 | 20.42 | 5.31 | 22.45 | -20.83 |
| 14000 | 3.80 | 19.55 | 4.55 | 21.11 | -16.43 |
| 16000 | 3.40 | 18.59 | 3.98 | 19.95 | -14.55 |
| 18000 | 3.00 | 17.50 | 3.54 | 18.93 | -15.18 |
| 20000 | 2.80 | 16.90 | 3.18 | 18.02 | -12.04 |
| 25000 | 2.40 | 15.56 | 2.55 | 16.08 | -5.75 |
| 30000 | 2.00 | 13.98 | 2.12 | 14.49 | -5.75 |
| 35000 | 1.80 | 13.06 | 1.82 | 13.16 | -1.04 |
| 40000 | 1.60 | 12.04 | 1.59 | 12.00 | 0.53 |
| 45000 | 1.40 | 10.88 | 1.41 | 10.97 | -1.04 |
| 50000 | 1.20 | 9.54 | 1.27 | 10.06 | -5.75 |
| 60000 | 1.00 | 7.96 | 1.06 | 8.47 | -5.75 |
| 70000 | 0.80 | 6.02 | 0.91 | 7.13 | -12.04 |
| 80000 | 0.80 | 6.02 | 0.80 | 5.97 | 0.53 |
| 90000 | 0.60 | 3.52 | 0.71 | 4.95 | -15.18 |
| 100000 | 0.60 | 3.52 | 0.64 | 4.04 | -5.75 |
| 200000 | 0.40 | 0.00 | 0.32 | -1.98 | 25.66 |
| 300000 | 0.20 | -6.02 | 0.21 | -5.51 | -5.75 |
| 400000 | 0.20 | -6.02 | 0.16 | -8.00 | 25.66 |
| 500000 | 0.20 | -6.02 | 0.13 | -9.94 | 57.08 |
| 600000 | 0.20 | -6.02 | 0.11 | -11.53 | 88.50 |
| 700000 | 0.20 | -6.02 | 0.09 | -12.87 | 119.91 |
| 800000 | 0.20 | -6.02 | 0.08 | -14.03 | 151.33 |
| 900000 | 0.20 | -6.02 | 0.07 | -15.05 | 182.74 |
| 1000000 | 0.20 | -6.02 | 0.06 | -15.96 | 214.16 |

## IV Bode plot




# 6 Exp. 3 Frequency Responses (II) Integrator with feedback resistor

## I Circuit

Measure frequency response of the circuit in Exp. 1 Integrator with the capacitor $C$ replaced with a capacitor $C = 0.001$ μF (102) and a resistor $R_F$ in parallel connection under

$$v_I = 0.2\sin(2\pi f t), \quad f = 100 \text{ Hz} \sim 1 \text{ MHz}$$

and sketch the Bode magnitude plot.

## II Theoretical results

Laplace domain:

$$V_+ = V_- = 0$$

$$I_I = \frac{V_I}{R} = I_F = -\left(\frac{1}{R_F} + Cs\right)V_O + Cv_O(0^-) = -\frac{1 + R_F Cs}{R_F}V_O + Cv_O(0^-)$$

$$V_O = \frac{R_F C}{1 + R_F Cs}v_O(0^-) - \frac{\frac{R_F}{R}}{1 + R_F Cs}V_I$$

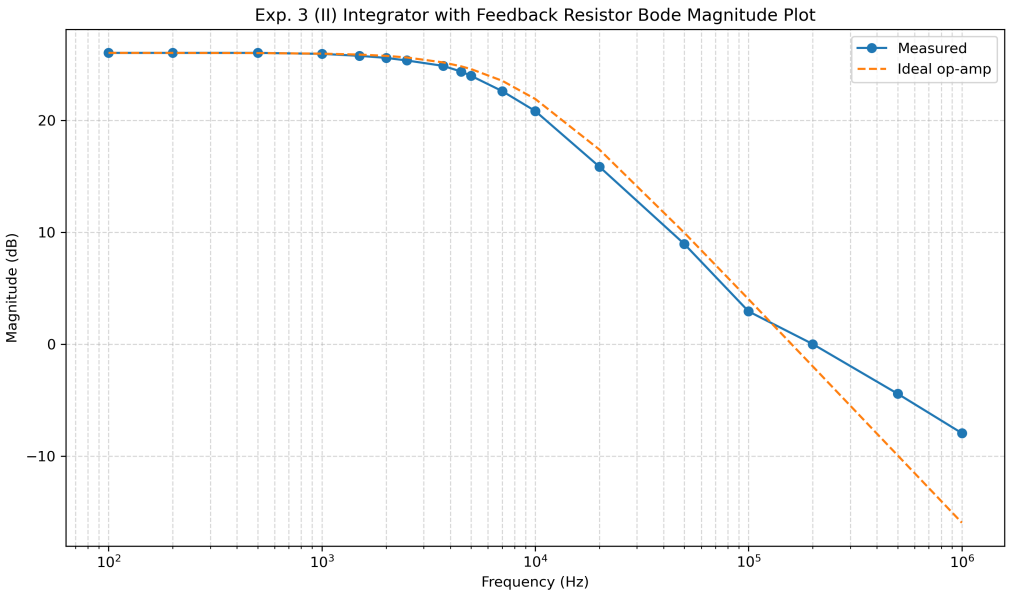
Transfer function: STC low-pass with cutoff frequency $\frac{1}{2\pi R_F C}$ and DC gain $\frac{R_F}{R}$.

$$H(s) = \frac{V_o}{V_i} = -\frac{\frac{R_F}{R}}{1 + R_F C s}$$

# III 實驗結果與數據分析

| frequency (Hz) | measured output voltage peak-to-peak (V) | measured gain (dB) | ideal output voltage peak-to-peak (V) | ideal gain (dB) | error w.r.t. ideal (%) |
|---|---|---|---|---|---|
| 100 | 8.00 | 26.02 | 8.00 | 26.02 | 0.01 |
| 200 | 8.00 | 26.02 | 8.00 | 26.02 | 0.03 |
| 500 | 8.00 | 26.02 | 7.98 | 26.00 | 0.20 |
| 1000 | 7.92 | 25.93 | 7.94 | 25.95 | -0.22 |
| 1500 | 7.76 | 25.76 | 7.86 | 25.87 | -1.29 |
| 2000 | 7.60 | 25.58 | 7.76 | 25.75 | -2.05 |
| 2500 | 7.40 | 25.34 | 7.63 | 25.61 | -3.04 |
| 3700 | 7.00 | 24.86 | 7.25 | 25.17 | -3.50 |
| 4500 | 6.60 | 24.35 | 6.96 | 24.82 | -5.22 |
| 5000 | 6.32 | 23.97 | 6.77 | 24.58 | -6.70 |
| 7000 | 5.40 | 22.61 | 6.01 | 23.53 | -10.10 |
| 10000 | 4.40 | 20.83 | 4.98 | 21.91 | -11.67 |
| 20000 | 2.48 | 15.85 | 2.96 | 17.38 | -16.15 |
| 50000 | 1.12 | 8.94 | 1.26 | 9.95 | -10.93 |
| 100000 | 0.56 | 2.92 | 0.63 | 4.01 | -11.76 |
| 200000 | 0.40 | 0.00 | 0.32 | -1.99 | 25.76 |
| 500000 | 0.24 | -4.44 | 0.13 | -9.94 | 88.52 |
| 1000000 | 0.16 | -7.96 | 0.06 | -15.96 | 151.34 |

# IV Bode plot



# 7　實驗心得